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Speech-Based ML for Cognitive Impairment Research

RESEARCH QUESTION

How do dataset quality, balance, and real-world recording conditions influence the performance of speech-based ML models?

DATASETS COMPARED

ADReSS

Standardised benchmark dataset

156 participants
4,077 speech segments
Balanced by age and gender, controlled settings

KI@Home

Home-collected speech dataset

79 participant households
4,793 speech segments
Heterogeneous, real-world conditions

WORKFLOW

Audio preprocessing → Feature extraction → Classification & regression → Cross-validation → Robustness evaluation

KEY TAKEAWAY

Model accuracy alone is not enough.
Reliable AI evaluation must consider dataset balance, data quality, recording conditions, and real-world variability.

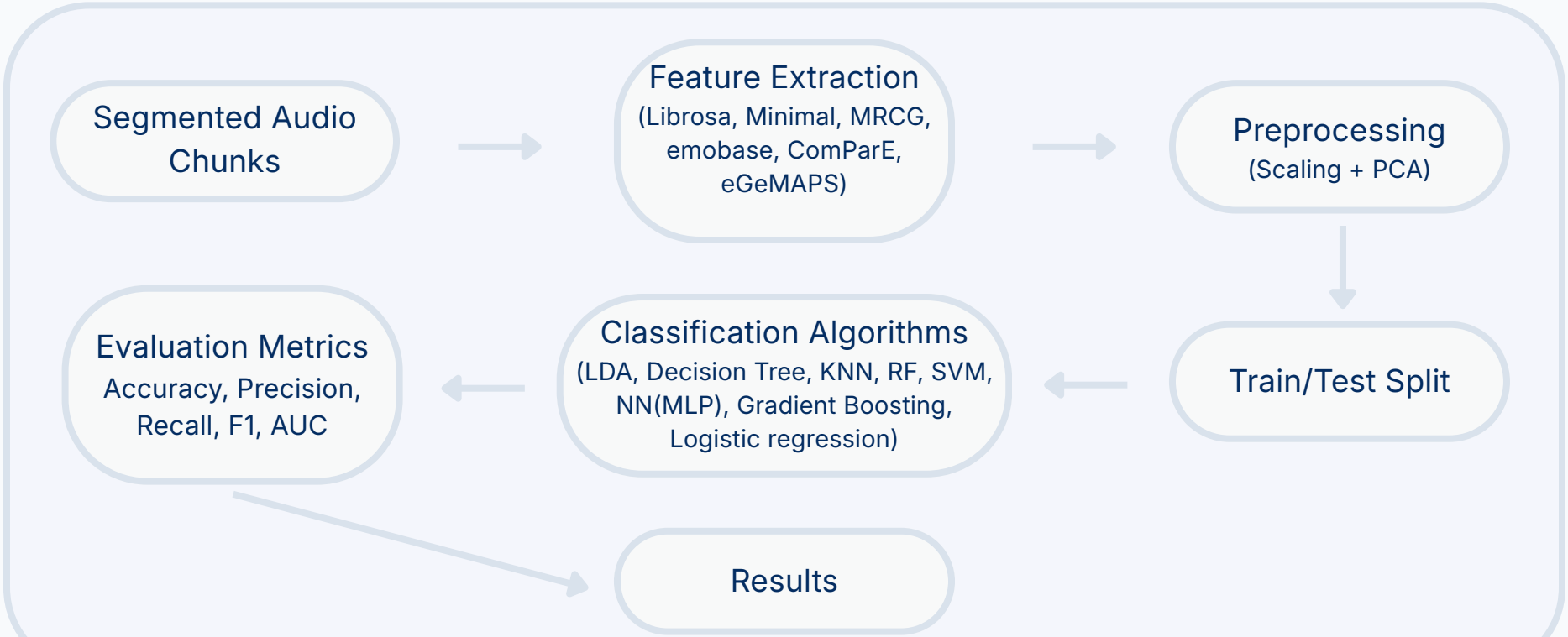
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Classification task

Speech segments were classified and aggregated through majority voting to produce subject-level results.

ARCHITECTURE DIAGRAM



Preprocessing → Acoustic features → Multiple classifiers → Segment-level prediction → Majority vote → Subject-level result

RESEARCH NOTE

Performance assessed using accuracy, precision, recall, F1 score, ROC-AUC, and cross-validation.

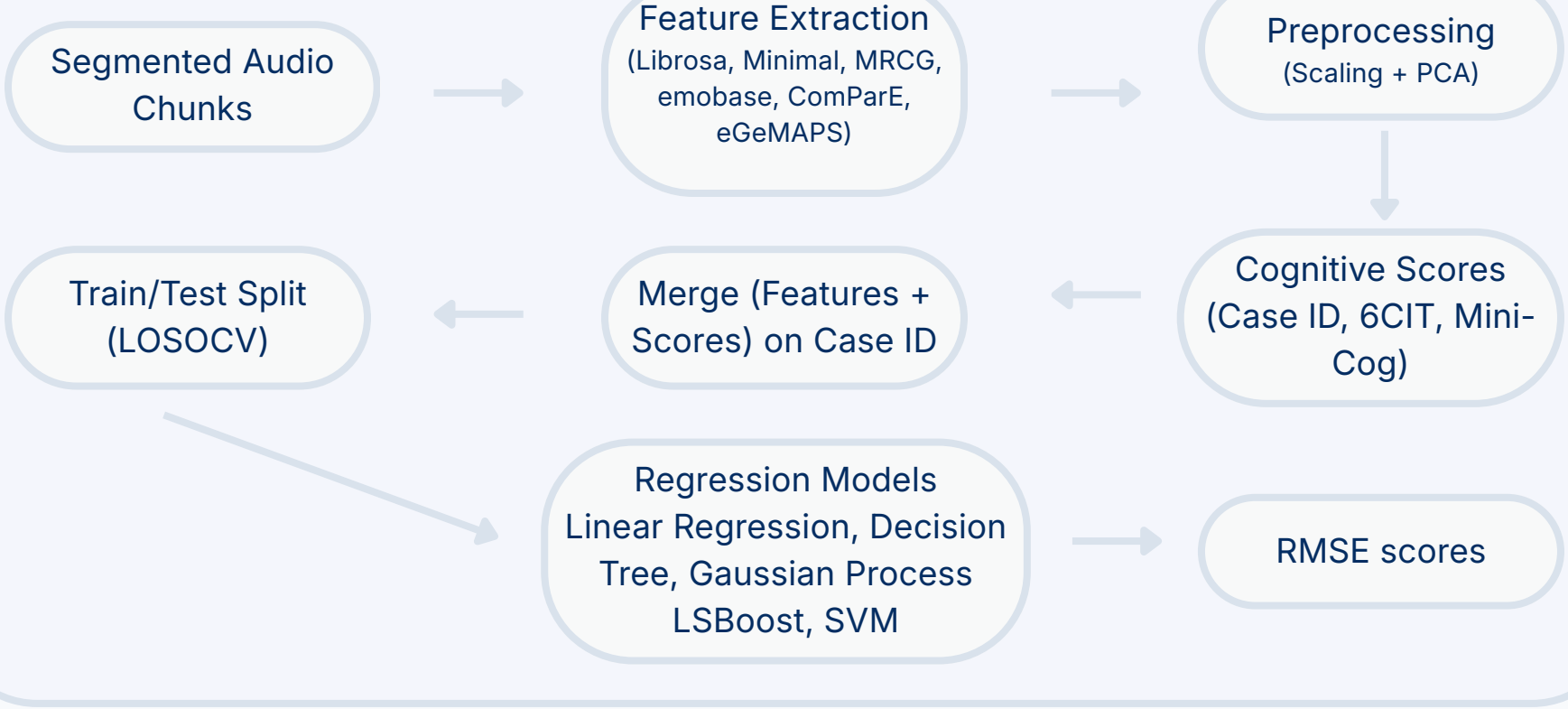
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Regression / Prediction Task

Speech-derived features were used to evaluate regression models for cognitive-screening score prediction.

ARCHITECTURE DIAGRAM



Preprocessing → Feature extraction → Regression models → Score prediction → Cross-validation and RMSE evaluation

RESEARCH NOTE

Mini-Cog score prediction was stable (RMSE 1–2); 6CIT prediction remained hard (RMSE 4–8, correlation near zero) — real-world speech captures only part of multi-domain cognition.

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Illustrative Best-Performing Models

The best-performing models from each dataset are presented as illustrative examples. Their performance should be interpreted in the context of the datasets' differing size, class balance, participant cohorts, and data-collection conditions.

KI@HOME

Neural Network MLP

0.8961	0.8804
Mean Accuracy	Mean F1 Score

ADReSS

Support Vector Machine

0.7271	0.7239
Mean Accuracy	Mean F1 Score

Results reflect two datasets with different cohort sizes, balance, and data-collection conditions; they should not be interpreted as clinical validation.